Building intelligent Global Health and
Humanitarian Response Technologies

Week 7:
The Fragility of Accuracy

Context How We Learned to *Count* What We Were Losing
Engineering When 99% Accuracy Is *Dangerous*

LiGHT Laboratory for intelligent Global Health & Humanitarian Response Technologies.

EPFL ARIADNE LABS ASHOKA UNIVERSITY WORLD ECONOMIC FORUM Rwanda Centre for the Fourth Industrial Revolution

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How We Learned to *Count* What We Were Losing

Lecturer: *Tim Arni*



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CONTEXT

Motivation

Central Question: How to correctly measure the performance of medical AI systems?

- Performance metrics **drive decision making** (research directions, fund allocations etc.)
- **Poorly performing models are misleading** and may lead to wrong clinical decisions
- Selecting **appropriate performance measures** and correctly interpreting them is essential

CONTEXT

Outline

01 ML perspective → More in Engineering Lecture

02 Medical AI perspective

03 Social sciences perspective

ML perspective

CONTEXT

ML is a (relatively) young research field and lacks strict reporting standards

Some important concepts:

A. Data leakage

B. Statistical significance of results

C. Evaluation of LLM outputs

D. Subgroup disparities

A) Data leakage

- **Data leakage** = model has during training (partial) **access to the data it is evaluated on** or test set data is used to select model configurations
- Split data into **train, val and test set** and correctly use the three data splits

Train set
fit model parameters and pre-processing decisions

Val set
tune hyperparameters, compare models, choose decision thresholds

Test set
use *once* for the final performance, claims only valid for that population

Data leakage

Types of Data leakage[1]:

1. **No clean separation of train and test data** (e.g. evaluate model on training data, pre-processing on training and test set, feature selection on training and test set, duplicates in train and test datasets)
2. **Illegitimate features**
3. Test set not drawn from the **distribution of scientific interest**

➔ Leads to **reproducibility failures and overoptimism**

➔ Specifically for LLMs: **decontamination** of training data (e.g. Apertus[2])

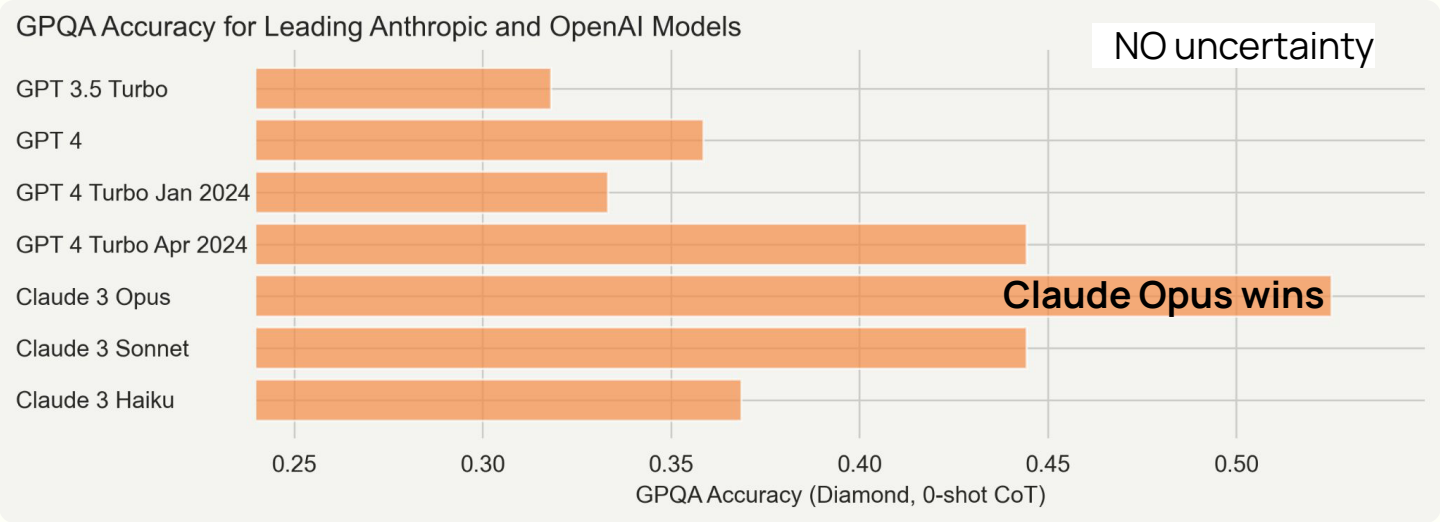
[1] Leakage and the Reproducibility Crisis in ML-based Science

[2] Project Apertus (2025). Democratizing Open and Compliant LLMs for Global Language Environments

B) Statistical significance of results

CONTEXT

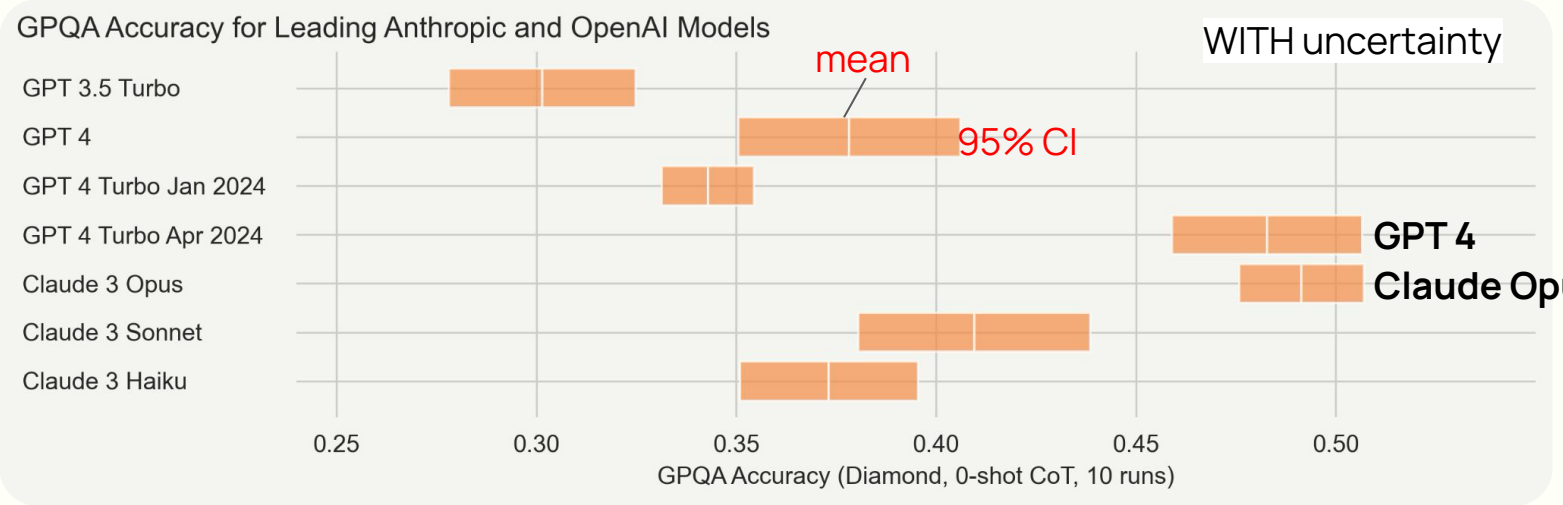
➔ Benchmark numbers drive decisions



Statistical significance of results

CONTEXT

➔ Without uncertainty estimates you cannot tell a real improvement from sampling noise (depending on sample size)



Statistical significance of results

CONTEXT

Some reasons for missing statistical significance of results:

- **Missing community guidelines**: Often little strict requirements
- **Social dynamics**[1]: ML research is heavily influenced by industry
- Potentially (very) **expensive** and time consuming
- Dataset size

[1] Biderman et al. (2026), Lessons from the Trenches on Reproducible Evaluation of Language Models

Statistical significance of results

CONTEXT

(Informal) Take-aways:

- Treat results from ML papers with some **skepticism**
- **Minimal best practice** (if possible):
  - Use a random seed for **reproducibility** (usually 42)
  - Run the same experiments **5 times with different train/test set splits** or at least run the evaluation 5 times on the same test dataset, using random seeds (usually 1 - 5)
  - **Report average and standard deviation** of the performance metric evaluated
  - Hint: Watch out for caching in API LLMs!
- If possible: do **appropriate statistical significance tests** (e.g. Welch's t-test)

C) Evaluation of LLM outputs

CONTEXT

- **Key Problem[1]:** a correct answer can be phrased in many semantically equivalent ways
- **Solutions**:
  - **Restrict answer space** (e.g. multiple-choice, string matching, verifiers etc.) → measurable, but less realistic
  - **Model-based grading** (BLEU/ROUGE score, LLM judges) → cheap and scalable, but some known flaws
  - **Human expert evaluation** → best but slow, expensive and can also be inconsistent
- **Scores are fragile[1]:** prompt wording, whitespace, few-shot examples and scoring method can significantly shift accuracy and change model rankings

[1] Biderman et al. (2026), Lessons from the Trenches on Reproducible Evaluation of Language Models

Evaluation of LLM outputs

CONTEXT

- Evaluating **multiple-choice answers vs. free text generation** can dramatically change the results
- Example: Changing evaluation protocol leads to big **differences in political values in LLM answers**

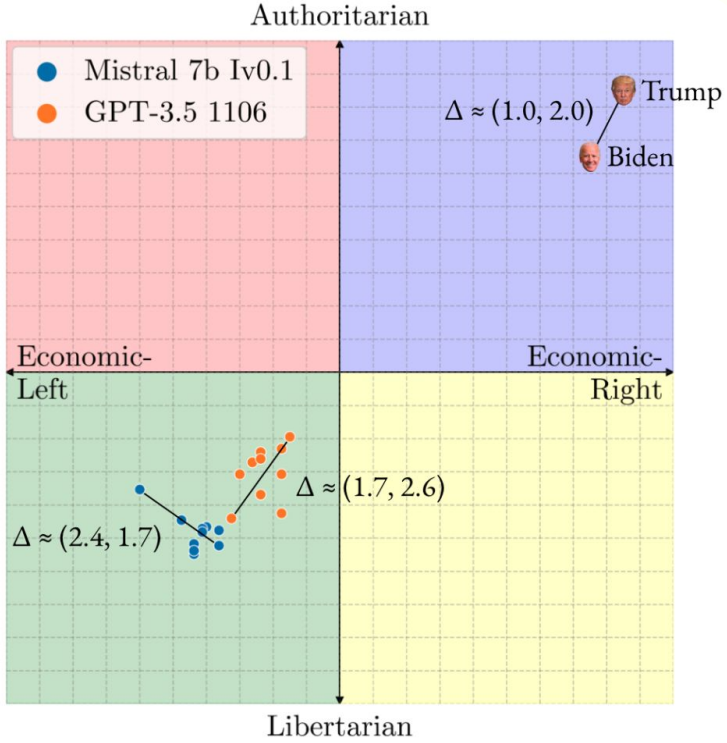


Figure from: Röttger et al. (2024), Political Compass or Spinning Arrow?

Evaluation of LLM outputs

CONTEXT

Recommendations[1] on evaluating genAI outputs:

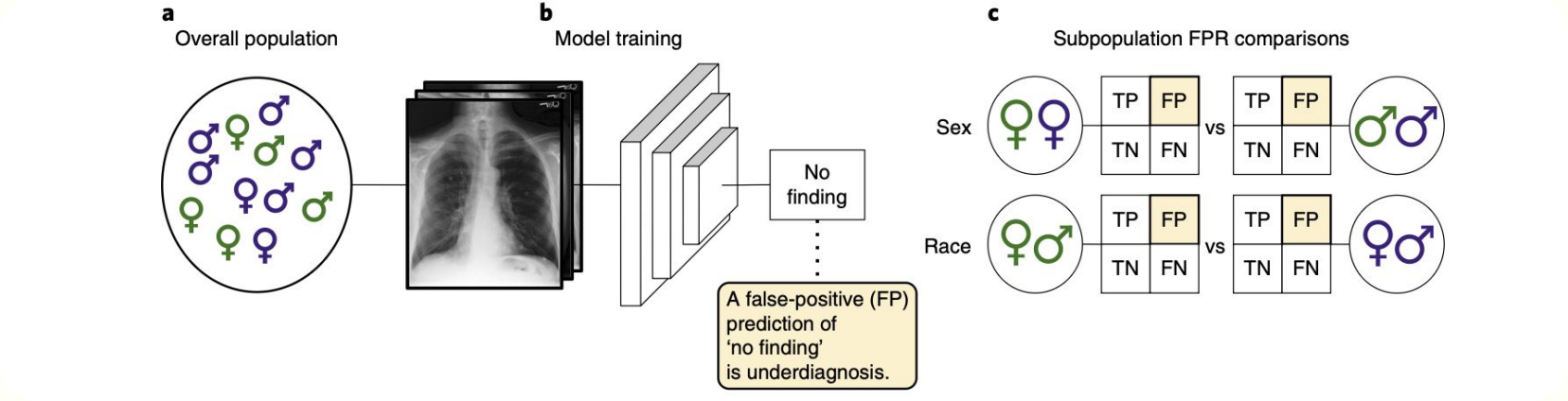
- **Match real use**: Test how users actually interact
- **Test robustness**: Vary prompts and evaluation protocols
- **Keep claims specific**: LLM A performs stronger than LLM B when evaluated following protocol Y on dataset Z

[1] and Figure from: Röttger et al. (2024), Political Compass or Spinning Arrow?

D) Subgroup disparities

CONTEXT

- **Subgroup disparities** = differences in performance against, or in favor of, a subpopulation



- Deep Learning model that **predicts no finding (positive)** or finding (negative)
- **False positive for "No finding" = a sick patient sent home as healthy** = underdiagnosis (worst type of error)

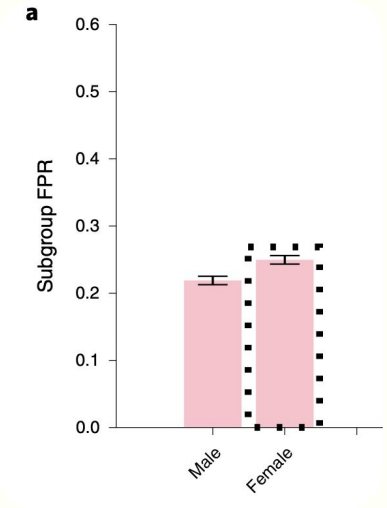
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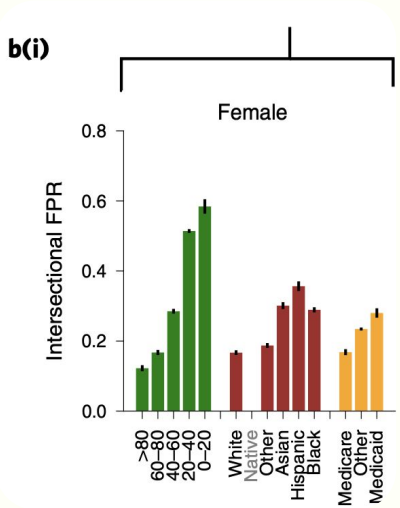
Figure from: Seyyed-Kalantari et al. (2021), Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in underserved patient populations

Subgroup disparities

CONTEXT

- FPR varies for different demographic subgroups
- Not just noise: FPR and FNR are inversely related across groups (in this example), so some groups are selectively labeled healthy
- **Intersectionality**: subgroups within subgroups
  - FPR higher for female than male patients
  - FPR higher for hispanic than white female patients





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Figure from: Seyyed-Kalantari et al. (2021), Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in underserved patient populations

Subgroup disparities

CONTEXT

- **Sources of bias:** Representation in data (55.13% male, 67.64% white where reported), using NLP methods to extract labels from radiology reports, clinicians underdiagnosed the same groups, universal threshold
- **Take aways**[1]:
  - Evaluate subgroup disparities during development and again after deployment (not just aggregate metrics)
  - Building fairer models is complicated and needs expert knowledge

[1] Seyyed-Kalantari et al. (2021), Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in underserved patient populations

Medical AI perspective

CONTEXT

Medicine is a mature research field and enforces strict standards

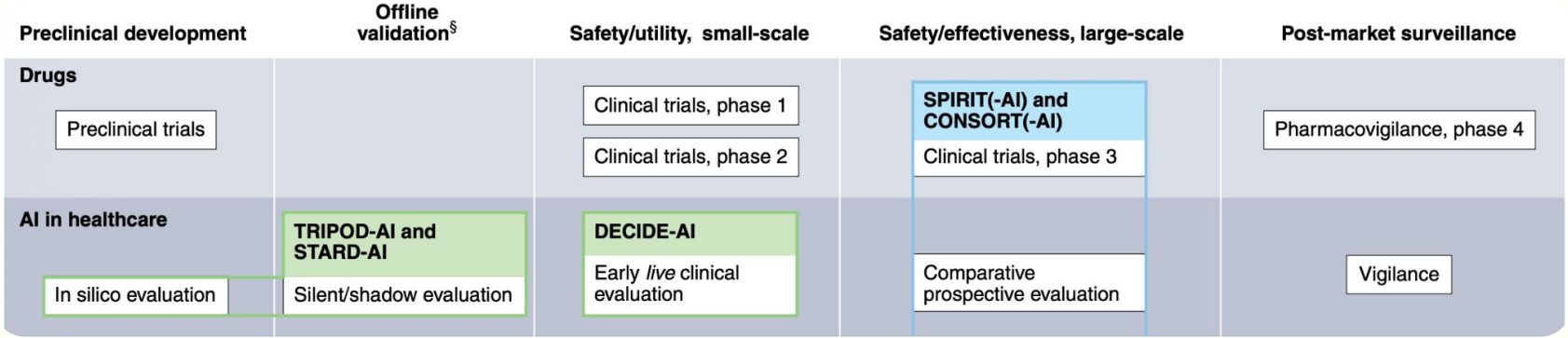
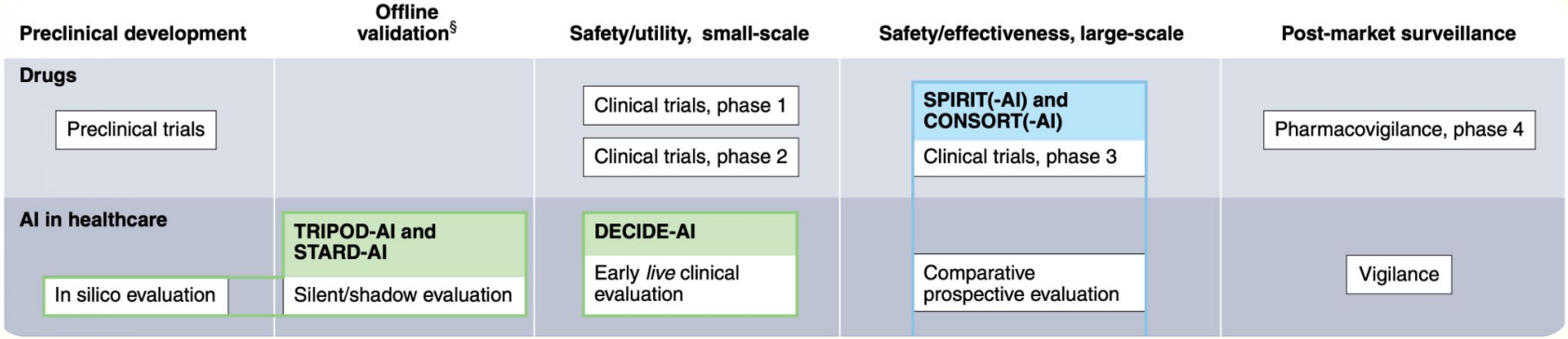


Figure from: Vasey et al. (2022). Nature medicine. Reporting guideline for the early-stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI

Medical AI perspective

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Medicine is a mature research field and enforces strict standards



Frameworks:

1. TRIPOD+AI and TRIPOD-LLM → model performance
2. STARD-AI → diagnostic accuracy
3. DECIDE-AI → human + AI system in a real workflow
4. SPIRIT/CONSORT-AI → randomized controlled trials of AI interventions (Week 12)

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Figure from: Vasey et al. (2022), Nature medicine, Reporting guideline for the early-stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI

Medical AI Reporting Guidelines

CONTEXT

Motivation:

- Systematic reviews show incomplete reporting, high risk of bias, overinterpreted results, etc.
- Avoid research waste → clear reporting allows replication, meta-reviews etc.

Medical AI Reporting Guidelines

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- Avoid research waste → clear reporting allows replication, meta-reviews etc.

Reporting guidelines:

- Defines the minimum set of information needed to ensure reproducibility, reduce bias, guarantee fairness etc.
- Created in a “Delphi process”
- Bridges medical and ML language: e.g. test set in ML is validation data in medical literature



Delphi process

“Delphi method process diagram” by Nesta, from Navigating policy complexity: a Delphi method approach, licensed under CC BY-NC-SA 4.0.

TRIPOD+AI

CONTEXT

- **T**ransparent **R**eporting of a multivariable prediction model for **I**ndividual **P**rognosis **O**r **D**iagnosis using **AI**
- Minimum reporting standard for studies that develop and/or evaluate a diagnostic or prognostic ML-based prediction model
- Detailed and concrete reporting checklists with heavy focus on bias reduction, fairness, open science etc.

TRIPOD+AI

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Table 2 | TRIPOD+AI checklist for the reporting of prediction model studies

Section/topic	Item	Development/evaluation*	Checklist item
Title			
Title	1	D;E	Identify the study as developing or evaluating the performance of a multivariable prediction model, the target population, and the outcome to be predicted
Abstract			
Abstract	2	D;E	See TRIPOD+AI for Abstracts checklist
Introduction			
Background	3a	D;E	Explain the healthcare context (including whether diagnostic or prognostic) and rationale for developing or evaluating the prediction model, including references to existing models
	3b	D;E	Describe the target population and the intended purpose of the prediction model in the context of the care pathway, including its intended users (eg, healthcare professionals, patients, public)
	3c	D;E	Describe any known health inequalities between sociodemographic groups
Objectives	4	D;E	Specify the study objectives, including whether the study describes the development or validation of a prediction model (or both)
Methods			
Data	5a	D;E	Describe the sources of data separately for the development and evaluation datasets (eg, randomised trial, cohort, routine care or registry data), the rationale for using these data, and representativeness of the data
	5b	D;E	Specify the dates of the collected participant data, including start and end of participant accrual; and, if applicable, end of follow-up
Participants	6a	D;E	Specify key elements of the study setting (eg, primary care, secondary care, general population) including the number and location of centres
	6b	D;E	Describe the eligibility criteria for study participants
	6c	D;E	Give details of any treatments received, and how they were handled during model development or evaluation, if relevant
Data preparation	7	D;E	Describe any data pre-processing and quality checking, including whether this was similar across relevant sociodemographic groups
Outcome	8a	D;E	Clearly define the outcome that is being predicted and the time horizon, including how and when assessed, the rationale for choosing this outcome, and whether the method of outcome assessment is consistent across sociodemographic groups
	8b	D;E	If outcome assessment requires subjective interpretation, describe the qualifications and demographic characteristics of the outcome assessors
	8c	D;E	Report any actions to blind assessment of the outcome to be predicted
Predictors	9a	D	Describe the choice of initial predictors (eg, literature, previous models, all available predictors) and any pre-selection of predictors before model building
	9b	D;E	Clearly define all predictors, including how and when they were measured (and any actions to blind assessment of predictors for the outcome and other predictors)
	9c	D;E	If predictor measurement requires subjective interpretation, describe the qualifications and demographic characteristics of the predictor assessors
Sample size	10	D;E	Explain how the study size was arrived at (separately for development and evaluation), and justify that the study size was sufficient to answer the research question. Include details of any sample size calculation

Table from: Collins et al. (2024), BMJ, TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods

TRIPOD+AI

CONTEXT

Methods		
Data	5a	D;E
	5b	D;E

Describe the sources of data separately for the development and evaluation datasets (eg, randomised trial, cohort, routine care or registry data), the rationale for using these data, and representativeness of the data
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Reporting guidelines in the ML field

CONTEXT

- NeurIPS Paper Checklist, Ethics statements, LLM Usage Statements etc. → are becoming more important but currently have limited coherence and impact
- **Model Cards** (Mitchell et al., 2019, Model Cards for Model Reporting) and **Datasheets for Datasets** (Gebru et al., 2021)
- **REFORMS** (Kapoor et al., 2024) → Structured checklist similar to the medical domain

➔ **Problem**: While medical ethics has a clear goal (*patient's health and wellbeing*) and long tradition this is not the case in AI ethics, complicating, among other reasons, the development of reporting guidelines[1]

[1] Mittelstadt (2019), Nature Machine Intelligence, Principles alone cannot guarantee ethical AI

Social Science perspective

CONTEXT

Social Sciences have thought a lot about how to measure (abstract) concepts

Measuring is complicated: lung cancer (easy) → depression → political values (hard, because an abstract concept)

Some researchers[1] argue that bias, fairness issues and reproducibility failures originate, among other reasons, in measurement mismatches (construct treated as its operationalization)

[1] Jacobs et al. (2021), Measurement and Fairness

Measurement modeling examples

CONTEXT

Example 1: Measure Height[1]

- **Concept:** Height
- **Operationalization**: Defining height as "*length from feet to top of head, standing erect*"
  - But: Do hair and shoes count? They matter for effective height in e.g. ergonomics
  - But: For someone in a wheelchair, sitting height might be the relevant quantity

[1]Example from: Jacobs et al. (2021), Measurement and Fairness

Measurement modeling examples

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  - But: Do hair and shoes count? They matter for effective height in e.g. ergonomics
  - But: For someone in a wheelchair, sitting height might be the relevant quantity
- Need **model for measurement error**:
  - Ruler measurements by doctors → assume errors follow a normal distribution is reasonable
  - Self-reported height data from dating apps → men tend to over-report their height (systematic error)

[1]Example from: Jacobs et al. (2021), Measurement and Fairness

Measurement modeling examples

CONTEXT

Example 2: Measure patient benefit[1] (cautionary tale)

Context: AI model used to enroll patients in a high-risk care management program

Concept: Patient benefit (unobservable)

[1]Examples from: Jacobs et al. (2021), Measurement and Fairness

Measurement modeling examples

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Concept: Patient benefit (unobservable)

Assumptions made by developers:

- Patient benefit → care needs → (future) care costs

Operationalization: predict care costs (using data from the past)

[1]Examples from: Jacobs et al. (2021), Measurement and Fairness

Measurement modeling examples

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Measurement error:

- Costs = need + past access to care
- **Important**: Even a perfect cost predictor does not solve the issue

[1]Examples from: Jacobs et al. (2021), Measurement and Fairness

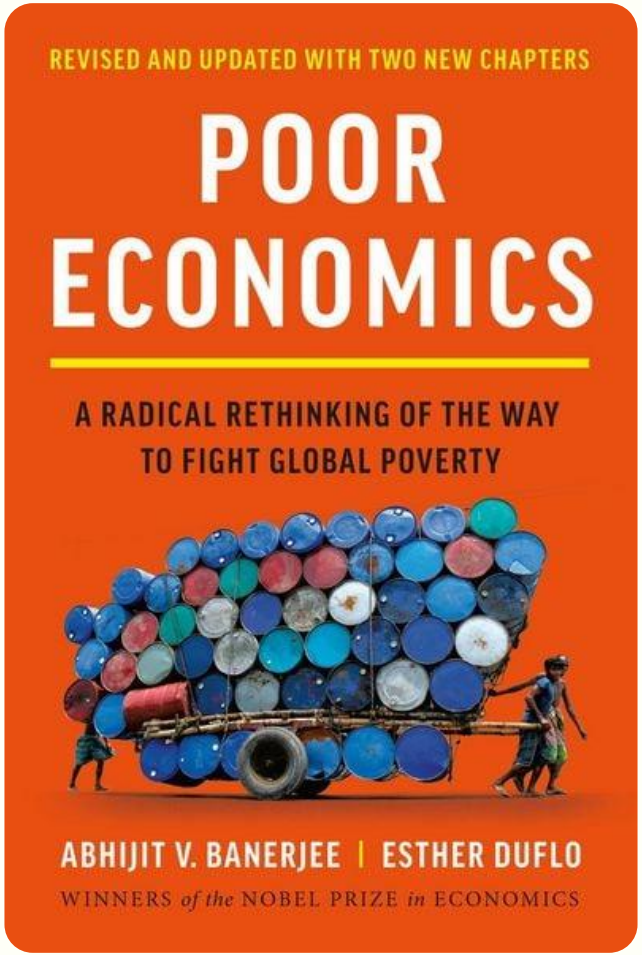
Social Science perspective

CONTEXT

Key messages:

- ➔ Take inspiration from other fields
- ➔ Include domain and cultural context experts
- ➔ Be open to critical reflection

Must reads



CONTEXT

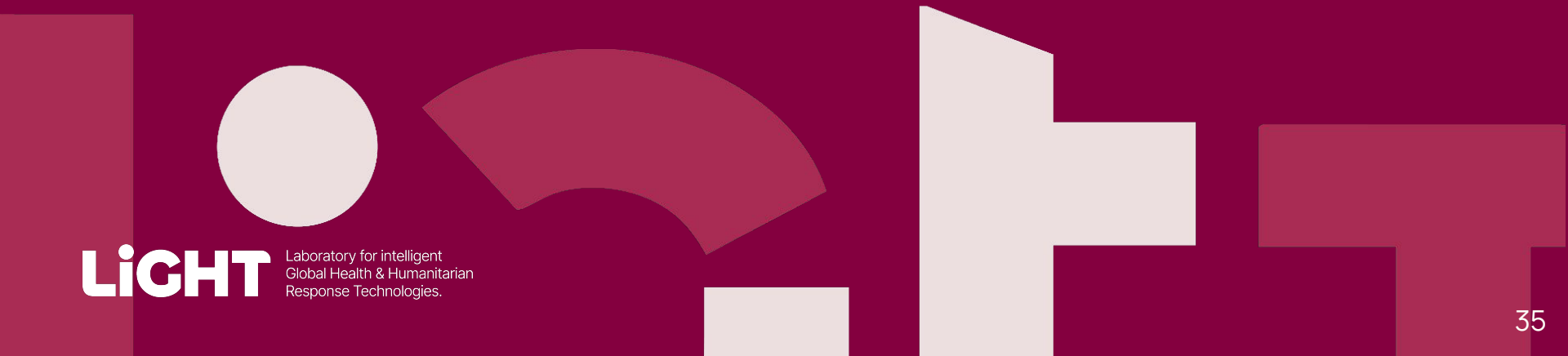
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Questions?

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When 99% Accuracy Is *Dangerous*

Lecturer: *Tim Arni*



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Outline

01 Recap of some important metrics

02 Discrimination and Calibration

03 Back to your Projects

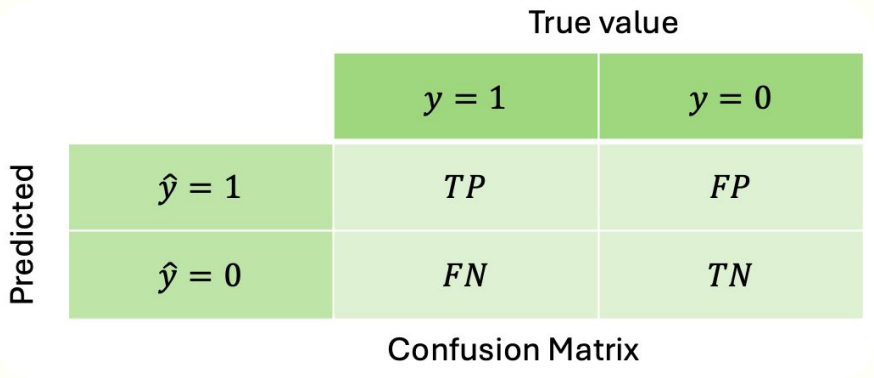
Ovarian Cancer Example

ENGINEERING

ADNEX prediction model:

- Malignant tumour (1) → advanced surgery (higher risk)
- Benign tumour (0) → conservative surgery

Problem: True diagnosis only known after surgery

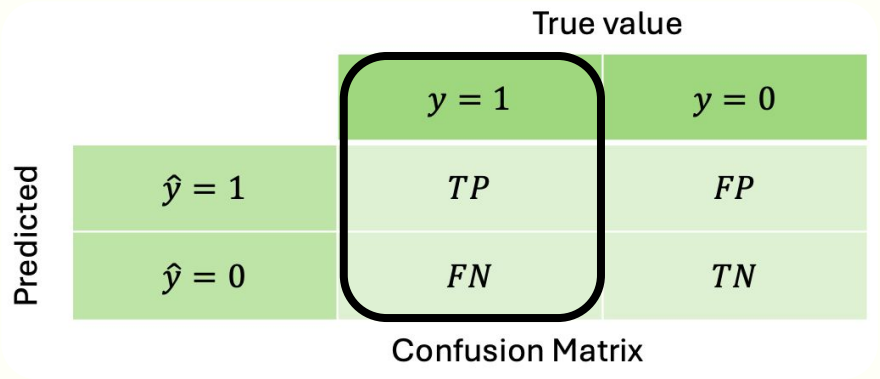


A) Recap on important metrics

ENGINEERING

Sensitivity or Recall:

- **Meaning**: "% of patients with cancer that were correctly classified"



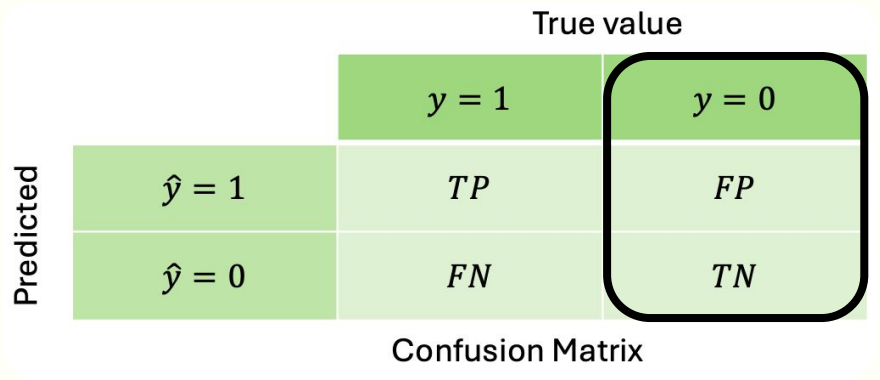
- Sensitivity $= \frac{TP}{TP+FN}$

Recap on important metrics

ENGINEERING

Specificity:

- **Meaning**: "% of patients without cancer that were correctly classified"



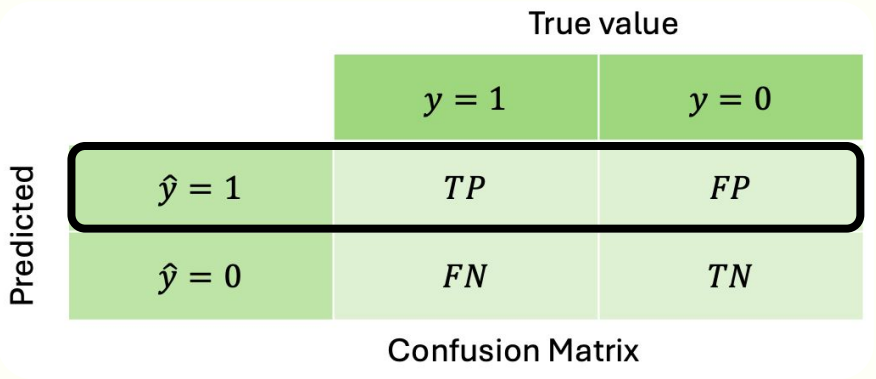
- Specificity $= \frac{TN}{TN+FP}$

Recap on important metrics

ENGINEERING

Precision or PPV:

- **Meaning**: “Of the patients the model flags positive, how many really have cancer?”



- $\text{Precision} = \frac{TP}{TP+FP}$

Recap on important metrics

ENGINEERING

Accuracy alone is not enough:

- For **low prevalence disease** (e.g. 1% of population affected)
- Evaluate dummy model that always predicts *label 0* on 10'000 patients
  - Accuracy = 99%
  - But: Sensitivity = 0%

- Sensitivity $= \frac{TP}{TP+FN}$

Recap on important metrics

ENGINEERING

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- Sensitivity $= \frac{TP}{TP+FN}$

Multi-class classification (with C classes):

- C x C confusion matrix
- E.g. Sensitivity per class c_i

Recap on important metrics

ENGINEERING

- Sensitivity etc. depend on threshold
- Compare two methods independent of the threshold
- **AUROC**
  - Sensitivity vs. 1 - Specificity
  - Probability that a randomly chosen patient with cancer receives a higher score than a randomly chosen patient without

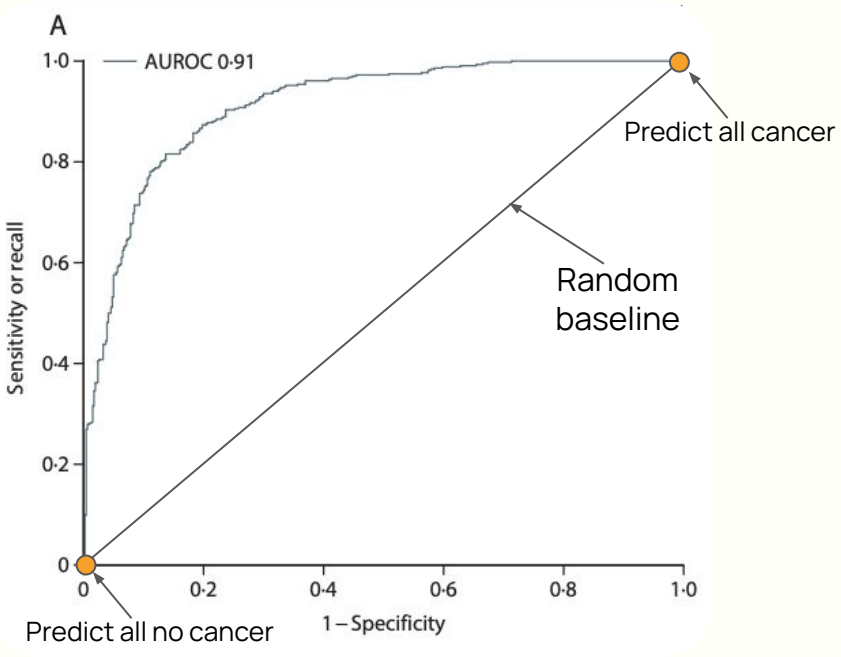
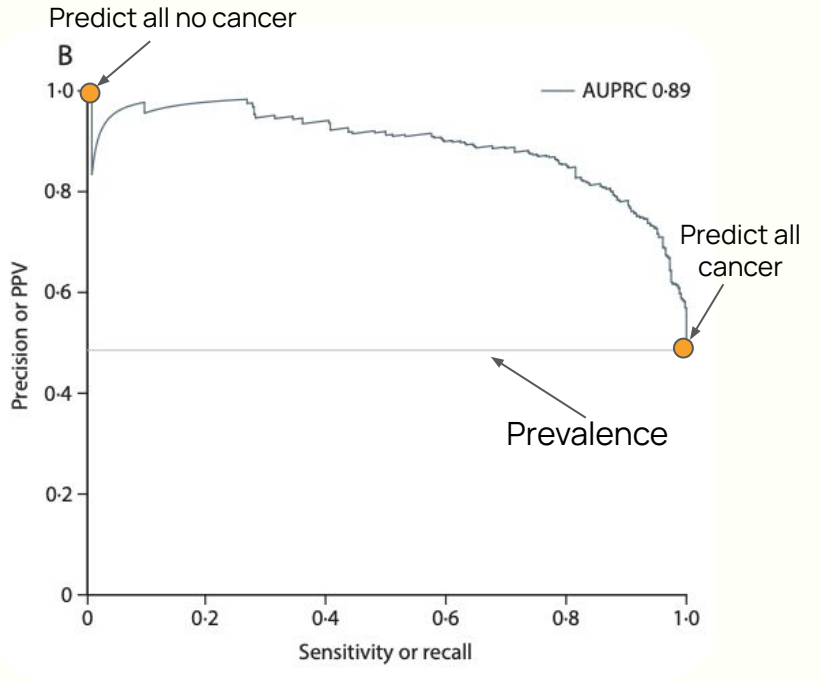


Figure from: Van Calster et al. (2025), Lancet Digit Health, Evaluation of performance measures in predictive artificial intelligence models to support medical decisions: overview and guidance

Recap on important metrics

ENGINEERING

- Sensitivity etc. depend on threshold
- Compare two methods independent of the threshold
- **AUPRC**
  - Precision vs. Sensitivity
  - Proportion of flagged patients that truly have cancer, averaged over all levels of sensitivity
  - Useful if low prevalence (does not depend on TN)



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Figure from: Van Calster et al. (2025), Lancet Digit Health, Evaluation of performance measures in predictive artificial intelligence models to support medical decisions: overview and guidance

B) Discrimination vs. Calibration

ENGINEERING

- Model predicts a risk for cancer (e.g. 0.1)
- **Discrimination**: "*Does the model rank patients correctly?*"
  - Patients with higher risks should get a higher score
  - Metrics: e.g. AUROC measures only discrimination[1]
- **Calibration**: "*Are the predicted probabilities correct?*"

[1] Van Calster et al. (2025), Lancet Digit Health, Evaluation of performance measures in predictive artificial intelligence models to support medical decisions: overview and guidance

Calibration

ENGINEERING

- **Confidence Calibration**: How well do the probabilities estimated by the model reflect real world likelihoods.
- Model predicts probability of cancer at 0.6 → 60% of these patients should have cancer

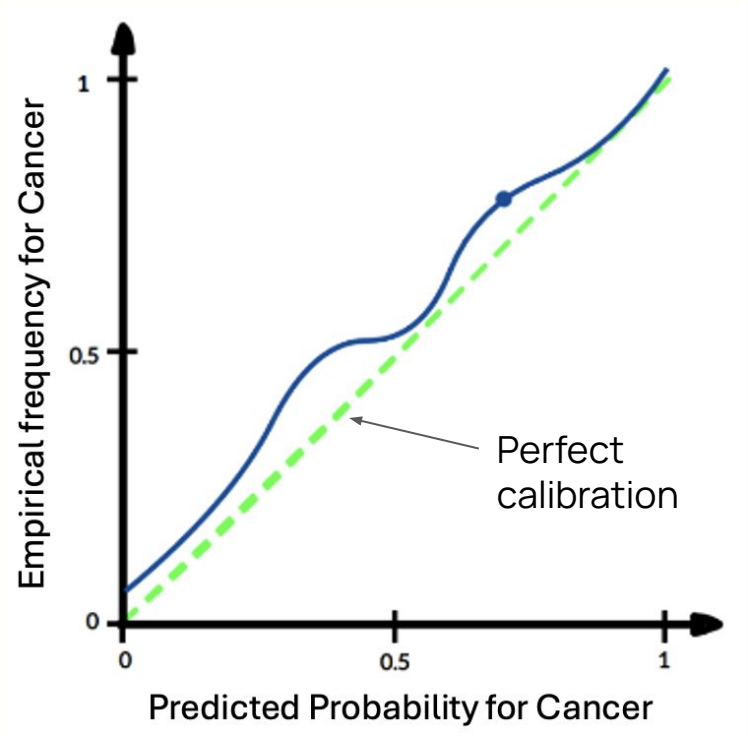
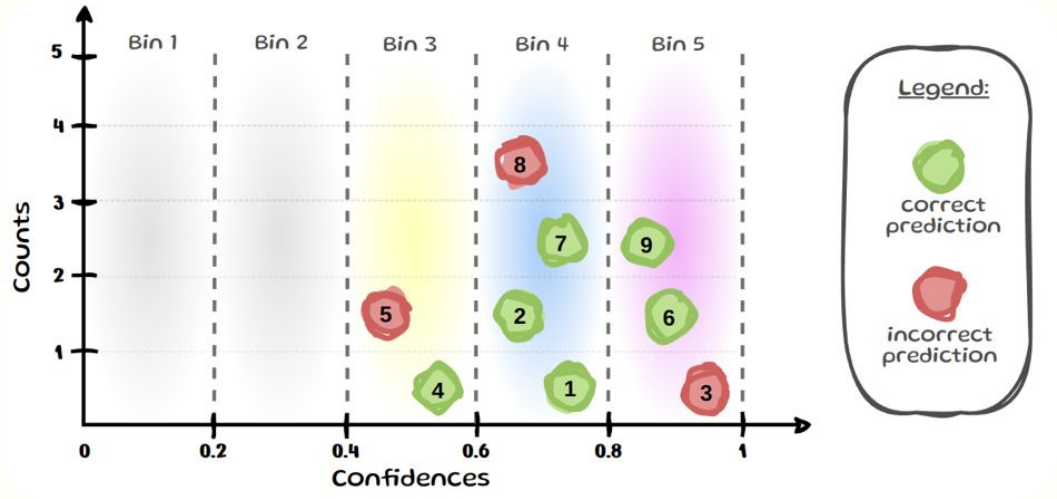


Figure adapted from: M. Pavlovic (2025). Understanding Model Calibration - A gentle introduction and visual exploration of calibration and the expected calibration error (ECE)

Calibration

ENGINEERING

- **Metric:** Expected Calibration Error (ECE)
- Split the n data points into M equally spaced bins



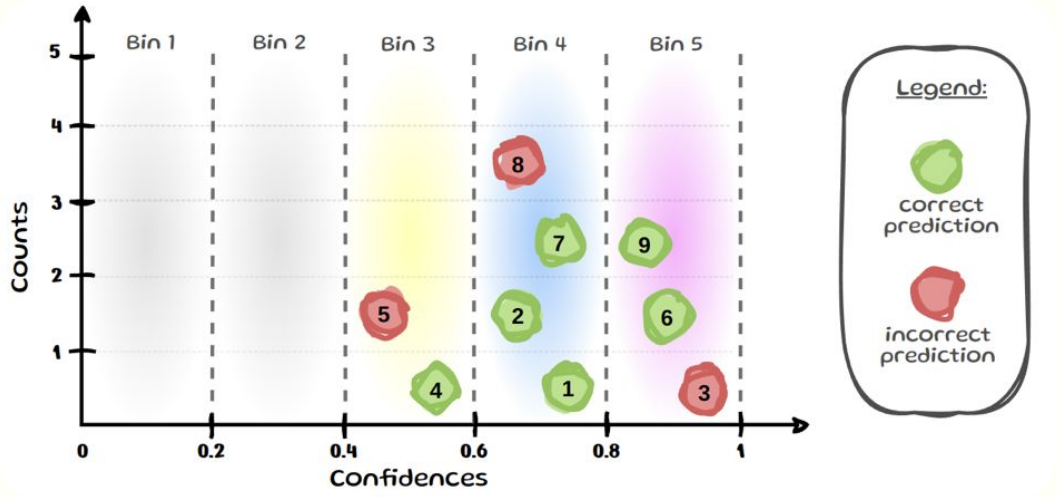
Figures from: M. Pavlovic (2025). Understanding Model Calibration - A gentle introduction and visual exploration of calibration and the expected calibration error (ECE)

Calibration

ENGINEERING

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- Split the n data points into M equally spaced bins

$$ECE = \sum_{m=1}^{M} \frac{|B_m|}{n} |acc(B_m) - conf(B_m)|$$



$$acc(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \mathbb{1}(\hat{y}_i = y_i)$$

$$conf(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \hat{p}(x_i)$$

Figures from: M. Pavlovic (2025). Understanding Model Calibration - A gentle introduction and visual exploration of calibration and the expected calibration error (ECE)

Calibration

ENGINEERING

- **Interpretation of ECE:**
  - Lower is better (ECE in [0, 1])
  - A perfectly calibrated model has ECE of 0
- **Drawbacks of ECE**:
  - Low ECE ≠ high accuracy
  - Number of bins influence value
- **Other metrics**[1]:
  - Adaptive ECE, Brier score etc.

[1] L. Lechuga et al. (2026). An Empirical Analysis of Calibration and Selective Prediction in Multimodal Clinical Condition Classification

Questions?

Building intelligent Global Health and Humanitarian Response Technologies

Thank You

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LiGHT Laboratory for intelligent Global Health & Humanitarian Response Technologies.

Our treat: a field trip to the ICRC Museum in Geneva, *covered by the lab*

Save the date !
November 14, the visit starts at **3PM**.



LiGHT Laboratory for intelligent Global Health & Humanitarian Response Technologies.